

# SambaNova RDU Platform Reference

Architecture, Compute Throughput, Memory Layout, Software Stack & Model Support

SambaNova Systems builds a full-stack AI platform — the Reconfigurable Dataflow Unit (RDU) chip, SambaRack/DataScale systems, and the SambaFlow/SambaStudio/SambaStack software layers — positioned as a dataflow-native alternative to GPUs for both training and, increasingly, large-scale **agentic inference**. This reference consolidates chip specifications, precision-level throughput, the three-tier memory architecture, the software/model stack, and the emerging Intel Xeon 6 heterogeneous-inference collaboration, for use alongside the Xeon 6 SKU workbook and storage/network/memory references.

Compiled Aug 2026 | Provenance-tagged: **I** Vendor-published **D** Derived/estimated **3P** Third-party analysis

## 1 Company & Product Snapshot

| Attribute                | Detail                                                                                                                                                                                                                                                            |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founded                  | 2017, Palo Alto, CA — Kunle Olukotun, Rodrigo Liang, Christopher Ré <b>I</b>                                                                                                                                                                                      |
| Core architecture        | Reconfigurable Dataflow Unit (RDU) — a coarse-grained reconfigurable array of Pattern Compute Units (PCU) and Pattern Memory Units (PMU), programmed via a dataflow graph compiled to a static execution plan (PEF), not SIMT/CUDA-style kernel dispatch <b>I</b> |
| Positioning              | Original focus: large-model training + inference (DataScale). Current focus (2026): purpose-built <b>agentic inference</b> — fast token generation, multi-model hot-swap, long context <b>I</b>                                                                   |
| System products          | DataScale (SN30-era), SambaRack (SN40L / SN50 rack systems), SambaCloud (hosted inference)                                                                                                                                                                        |
| Funding / scale          | >\$1.48B raised historically; \$350M+ Series E (Feb 2026) with Vista Equity Partners, Battery Ventures, T. Rowe Price accounts, valuation ~\$5B+ (2021 mark) <b>I</b>                                                                                             |
| Strategic partner (2026) | Intel — multi-year collaboration: Xeon 6 as host/action CPU + RDU for decode, joint go-to-market; SoftBank is first SN50 deployment customer (Japan) <b>I</b>                                                                                                     |

## 2 RDU Generations at a Glance

| Generation      | Codename      | Process                                 | Transistors | On-chip SRAM            | HBM         | DDR (attached)                      | Peak BF16/FP16 TFLOPS (per socket)          |
|-----------------|---------------|-----------------------------------------|-------------|-------------------------|-------------|-------------------------------------|---------------------------------------------|
| SN10 (Gen 1)    | Cardinal SN10 | 7nm TSMC                                | —           | 320 MB                  | None        | 1.5 TB DDR4 (system-level)          | 320 TFLOPS BF16 <b>3P</b>                   |
| SN30 (Gen 2)    | Cardinal SN30 | 7nm TSMC                                | 86B         | 640 MB                  | None        | 1 TB direct-attached DDR            | 688 TFLOPS BF16 <b>I</b>                    |
| SN40L (Gen 3/4) | Cerulean      | 5nm TSMC (CoWoS-S, dual die)            | 102B        | 520 MiB                 | 64 GiB HBM3 | Up to 1.5 TiB DDR (pluggable DIMMs) | 638–640 TFLOPS BF16 <b>I</b>                |
| SN50 (Gen 5)    | Cerulean 2    | ~3nm TSMC (est., dual chiplet) <b>D</b> | —           | 432 MB (216 MB/chiplet) | 64 GB HBM2E | Up to 2 TB DDR5 <b>3P</b>           | ~1.6 PFLOPS BF16/FP16 (2.5× SN40L) <b>D</b> |

**Reading the SN50 row:** SambaNova has not published absolute per-socket TFLOPS/PFLOPS for SN50 in primary marketing collateral — official claims are stated as *relative* multiples versus SN40L (5× compute, 4× network bandwidth, 2.5× BF16 throughput, FP8

support at 2× that). The ~1.6 PFLOPS BF16 / ~3.2 PFLOPS FP8 absolute figures below come from independent technical-press derivation (Next Platform, industry newsletter analysis) applying those multipliers to the known SN40L baseline — treat as estimates pending an official spec sheet.

### 3 Compute Throughput by Data Type (TFLOPS)

#### SN40L (current-generation, GA)

| Data type                           | Per-socket TFLOPS           | Notes                                                                                                                                                                               |
|-------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BF16 / FP32 (native)                | 638–640 TFLOPS              | Native hardware datapath; PCU SIMD stage supports FP32, BF16, INT32 natively. Vendor blog cites a fleet roofline of 10.2 PFLOPS BF16 across a 16-socket SambaRack SN40L-16 <b>I</b> |
| FP8                                 | Not a native SN40L datapath | SN40L generation does not add dedicated FP8 throughput doubling — FP8 support arrives with SN50 <b>D</b>                                                                            |
| INT8 / INT32                        | Supported (SIMD ALU)        | Used for quantized inference paths; PCU configurable as INT32 SIMD lanes <b>I</b>                                                                                                   |
| 16-socket rack (SambaRack SN40L-16) | 10.2 PFLOPS BF16 aggregate  | Peer-to-peer interconnect across 16 sockets in one SambaRack <b>I</b>                                                                                                               |

#### SN50 (ships H2 2026)

| Data type                | Per-socket (est.)                                 | Notes                                                                                                                           |
|--------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| BF16 / FP16              | ~1.6 PFLOPS                                       | 2.5× SN40L BF16 throughput per vendor relative claim <b>D</b>                                                                   |
| FP8 (new, native)        | ~3.2 PFLOPS                                       | New native FP8 datapath — 2× the BF16 rate; first RDU generation with dedicated FP8 <b>D</b>                                    |
| FP4                      | No additional FLOPS uplift                        | SN50 can represent FP4 data (memory-footprint benefit) but executes it through the FP8 datapath — no extra throughput <b>3P</b> |
| 256-chip scale-up domain | Model classes >10T parameters, >10M token context | Max single-inference-worker model-parallel span is 256 SN50 sockets; broader Ethernet scale-out domain up to 32K RDUs <b>I</b>  |

**Benchmark comparison vs. NVIDIA B200 (SemiAnalysis InferenceX, cited by SambaNova):** Llama 3.3 70B at FP8, 1K/1K tokens — B200 max speed 184 tok/s/user vs. SN50 895 tok/s/user (~5×). SN50 also claims >3× aggregate throughput per chip vs. B200. Raw peak FLOPS still trail B200 (SN50 ~3.2 PFLOPS FP8 vs. B200 ~5.0 PFLOPS FP8 est.) — SambaNova's advantage is architectural (dataflow keeps agentic decode/KV-cache traffic on-chip) rather than raw peak compute. **I 3P**

### 4 Three-Tier Memory Architecture

Every RDU generation since SN40L uses the same conceptual three-tier hierarchy: on-chip SRAM (PMU scratchpad) → on-package HBM → off-package DDR DRAM. The dataflow compiler places tensors across tiers based on access pattern, not a fixed cache hierarchy — this is the architectural basis for SambaNova's "resident multi-model" and agentic KV-cache claims.

| Tier                        | SN40L capacity                                        | SN50 capacity                                                                 | Role                                                                                                                                  |
|-----------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Tier 0 — On-chip SRAM (PMU) | 520 MiB, hundreds of TBps aggregate on-chip bandwidth | 432 MB (216 MB × 2 chiplets) — capacity traded for core-count increase        | Fastest tier; activations, hot weights, streaming intermediate tensors; concurrent read/write scratchpad with 4 addr/cycle generation |
| Tier 1 — On-package HBM     | 64 GiB HBM3                                           | 64 GB HBM2E, ~1.84 TB/s <b>3P</b><br>(unchanged capacity; step back to HBM2E) | Router/expert weights in active use; agentic KV-cache tier; input-token caching to cut TTFT                                           |

generation, roughly flat bandwidth vs. SN40L)

|                                           |                                                          |                                                        |                                                                                                                                          |
|-------------------------------------------|----------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Tier 2 — Off-package DDR (attached DIMMs) | Up to 1.5 TiB DDR, >1 TB/s load rate<br>DDR→HBM per node | Up to 2 TB DDR5 <b>3P</b> (generation moves DDR4→DDR5) | Cold/warm weight residency for large MoE / CoE expert pools; enables >1T-parameter models to stay chip-resident without host round-trips |
|-------------------------------------------|----------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|

**Why this matters for Xeon 6 capacity planning:** SN40L/SN50 keep model weights on the accelerator card across all three tiers — there is *no PCIe round-trip to host DRAM* for weights during inference, unlike a GPU pulling from host memory on cache miss. The Xeon 6 host in a heterogeneous SambaNova deployment is sized for orchestration, prefill hand-off, and "action" workloads (tool calls, agent control-plane), not for streaming model weights — this changes the CPU-memory-bandwidth sizing calculus versus a GPU-centric node.

Node / Rack Reference Configurations

| System                          | RDU count                             | Aggregate HBM       | Aggregate DDR                             | Host CPU                                      | Power                                                               |
|---------------------------------|---------------------------------------|---------------------|-------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------|
| DataScale SN40L (8-socket node) | 8× SN40L                              | 512 GiB             | up to 12 TB                               | 2× 64-core, 2 TB DDR4                         | —                                                                   |
| SambaRack SN40L-16              | 16× SN40L                             | 1 TB                | 12 TB                                     | 2× 64-core, 2 TB DDR4                         | Inference: 7–14.5 kW (typical 10 kW)                                |
| SambaRack SN50                  | 16× SN50 (scales to 256 across racks) | 1 TB (16-chip rack) | up to 32 TB (16-chip rack, est.) <b>D</b> | Intel Xeon 6 (new, per Intel collab) <b>I</b> | ~20 kW/rack, air-cooled, no liquid-cooling retrofit needed <b>I</b> |

## 5 Software Stack

| Layer                     | Component                         | What it does                                                                                                                                                                                                                                                                                           |
|---------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compiler / Runtime        | SambaFlow                         | Core software stack running on DataScale/SambaRack hardware. Traces a submitted PyTorch model graph, compiles it to a PEF (dataflow execution file), and runs it for training or inference. Supports Ubuntu 22.04.x and Red Hat 8.8; packages standardized under <code>/opt/sambanova/</code> <b>I</b> |
|                           | SambaNova Runtime                 | Low-level execution/scheduling layer beneath SambaFlow; versioned independently (e.g., 1.22, 1.24) <b>I</b>                                                                                                                                                                                            |
| Model lifecycle           | SambaStudio                       | GUI-based platform for the full model lifecycle: train, fine-tune with proprietary data, deploy to production endpoints, manage workloads. CLI access via <code>snap</code> ; SDK via <code>snsdk</code> <b>I</b>                                                                                      |
|                           | SambaTune                         | Performance analysis/tracing tool — traces a PyTorch operator or "hyperfunction" across stack layers (Stack Tracing report) <b>I</b>                                                                                                                                                                   |
| Enterprise platform       | SambaStack                        | Full-stack enterprise inference platform (successor branding to DataScale/SambaStudio for production): deploy/manage/scale models on-prem or dedicated cloud, powered by SambaRack hardware. Supports rapid model hot-swap for agentic workflows on a single node <b>I</b>                             |
| Hosted inference          | SambaCloud                        | Hosted, full-stack inference platform; OpenAI-compatible API (set <code>OPENAI_API_KEY</code> to SambaNova key + base URL); integrates with CrewAI, Hugging Face, Cline, AWS <b>I</b>                                                                                                                  |
| Compiler (inference path) | SambaNova Composer                | Proprietary layer that compiles model graphs for the RDU dataflow execution model; only supports architectures it can map — no vLLM/SGLang/custom-CUDA-kernel compatibility <b>3P</b>                                                                                                                  |
| Fault management          | SambaNova Fault Management (SNFM) | Reporting, diagnosis, and analysis of system error/fault events for a DataScale/SambaRack deployment <b>I</b>                                                                                                                                                                                          |

### Host OS / Integration Notes

| Item                        | Detail                                                                                                                                       |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Supported OS                | Ubuntu 22.04.x, Red Hat Enterprise Linux 8.8 <b>I</b>                                                                                        |
| Model framework in          | PyTorch (model definition, graph tracing) — no native TensorFlow/ONNX front end documented                                                   |
| API compatibility           | OpenAI-compatible REST endpoints on SambaCloud — drop-in for OpenAI SDK clients <b>I</b>                                                     |
| Ecosystem integrations      | CrewAI, Hugging Face, Cline, AWS (per SambaCloud integration list) <b>I</b>                                                                  |
| GPU interoperability (2026) | Heterogeneous inference: NVIDIA H200 GPUs for prefill + SN50 RDUs for decode, orchestrated across the stack (disaggregated serving) <b>I</b> |

## 6 Model Support

### 6.1 Composition of Experts (CoE) — SambaNova's native model architecture

CoE composes many independently-trained expert models behind a router, versus a Mixture-of-Experts (MoE) which trains a single sparse network end-to-end. CoE experts can be heterogeneous in architecture and size and are trained/fine-tuned independently — the RDU's three-tier memory (weights resident in DDR, router in HBM, active expert paged to HBM) is what makes serving 50+ experts from one node practical. **I**

| Model                       | Scale                       | Composition                                                                                                                                            | Deployment footprint                                                                                    |
|-----------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Samba-1                     | 1.3T parameters (aggregate) | 56 component models, 4 model architectures, 10 domains, 30+ languages; includes Llama 2, Mistral, DeepSeek Coder, Falcon, DePlot, CLIP, LLaVA <b>I</b> | 8 RDU sockets (1 SN40L node) for the full composition <b>I</b>                                          |
| Samba-CoE (research config) | >1T parameters              | 150 Llama2-7B-derived experts + router                                                                                                                 | 1 SN40L node (8 sockets); 19× footprint reduction, 15–31× faster model-switch vs. DGX baseline <b>I</b> |

6.2 Third-party open models served (SambaCloud / SambaStack, 2026)

| Model family   | Notable variants live                                                                                | Notes                                                                               |
|----------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Meta Llama     | Llama 3.1 (8B/70B/405B — first platform to serve all three), Llama 3.3 70B, Llama 4 Scout & Maverick | Launch partner for Llama 4 <b>I</b>                                                 |
| DeepSeek       | DeepSeek-V3.1, DeepSeek R1 671B                                                                      | DeepSeek-V3.1 up to ~200 tok/s (Artificial Analysis measured) <b>I</b>              |
| OpenAI gpt-oss | gpt-oss-120b                                                                                         | >600–700 tok/s on RDU — MoE architecture suits dataflow expert-paging well <b>I</b> |
| MiniMax        | MiniMax-M2.7                                                                                         | Highest tracked intelligence-index model on the platform as of mid-2026 <b>3P</b>   |
| Google Gemma   | Gemma 4 31B                                                                                          |                                                                                     |
| Qwen           | Qwen family (platform page)                                                                          |                                                                                     |

Modalities: text, image, and audio processing are supported depending on model (SambaCloud platform description). Pricing/throughput figures move quickly — treat the table above as a snapshot, not a live catalog. **3P**

7 Intel Xeon 6 Heterogeneous Inference Collaboration

Announced Feb 24, 2026 alongside SN50 and reaffirmed at Computex (Jun 2, 2026) with Foxconn as manufacturing partner. This is the most directly relevant thread for Xeon 6 capacity planning.

| Element               | Detail                                                                                                                                                                                                                              |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Architecture          | GPU (NVIDIA H200) handles compute-heavy <b>prefill</b> ; Xeon 6 serves as host <i>and</i> "action" CPU (agent tool-call / control-plane execution); SN50 RDU handles memory-bandwidth-bound <b>decode</b> <b>I</b>                  |
| Host CPU role change  | Intel Xeon 6 replaces AMD EPYC as SambaNova's primary host CPU platform going forward (multi-year partnership) <b>3P</b>                                                                                                            |
| Manufacturing partner | Foxconn — rack-scale production systems <b>I</b>                                                                                                                                                                                    |
| Demonstrated result   | 4× NVIDIA H200 (prefill) + 16× SN50 in one SambaRack (decode): 763 tok/s on MiniMax-M2.7 at 10K-token context (Artificial Analysis-measured, reported by The Register, Jul 8 2026); >450 tok/s sustained at longer context <b>I</b> |
| Availability          | Joint solution targeted for enterprise/cloud/sovereign AI customers in H2 2026; SN50 customer shipments begin H2 2026 <b>I</b>                                                                                                      |
| First deployment      | SoftBank Corp., next-generation AI data centers in Japan <b>I</b>                                                                                                                                                                   |
| Status caveat         | As of the Jun 2026 Computex announcement, this was described as an "intent to build production rack-scale systems" — a multi-year collaboration, not a shipping SKU with published pricing <b>3P</b>                                |

**Workbook cross-reference:** This disaggregated prefill/decode pattern implies two distinct Xeon 6 host roles in a SambaNova-attached node — (a) a lighter host/orchestration CPU per SN50 RDU rack (memory bandwidth for KV-cache staging and agent control-plane, not bulk weight streaming), and (b) potentially a separate Xeon 6 prefill-adjacent host if GPUs are Xeon-hosted in the same pod. Suggested as a candidate new row class in the GenAI Infrastructure tab once official Xeon 6 SKU pairings are published by Intel/SambaNova.

## 8 Sources & Assumptions

| Tag | Meaning                                                                                                                                                                  | Applied to                                                                                                                                                                             |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I   | Directly vendor-published (SambaNova blog/docs/press release, IEEE/Hot Chips paper co-authored by SambaNova, business-wire release)                                      | SN40L specs, SN30 specs (EE Times/HPCwire citing vendor), Samba-1 composition, SambaFlow/SambaStudio/SambaStack docs, Intel partnership announcement, SN50 relative-performance claims |
| D   | Derived — a figure calculated by applying a vendor-stated multiplier or ratio to a known published baseline, not itself published as an absolute number                  | SN50 absolute PFLOPS (BF16/FP8) from the "2.5× / FP8 at 2× that" multiplier on the SN40L 640-TFLOPS baseline; SN50 process node                                                        |
| 3P  | Independent technical press / analyst estimate not confirmed by a SambaNova-published spec sheet (Next Platform, SemiAnalysis InferenceX, Spheron, industry newsletters) | SN50 HBM bandwidth (~1.84 TB/s), SN50 SRAM/DDR figures, SN40L effective HBM bandwidth (~2 TB/s), B200 FP8 comparison figure                                                            |

Primary sources consulted: [sambanova.ai](https://sambanova.ai) (product pages, blog posts Feb–Jul 2026), [docs.sambanova.ai / docs-legacy.sambanova.ai](https://docs.sambanova.ai/docs-legacy.sambanova.ai), [arXiv 2405.07518](https://arxiv.org/abs/2405.07518) (SN40L Composition-of-Experts paper), IEEE Xplore 16.4 SN40L ISSCC paper, Hot Chips 2024 SambaNova presentation (Prabhakar et al.), [businesswire.com](https://businesswire.com) SN50 launch release (Feb 24, 2026), Intel Capital press coverage, EE Times / HPCwire / The Next Platform / The Register (SN10/SN30/SN50 hardware analysis), Artificial Analysis provider page, Tom's Hardware, ALCF AI Testbed hardware listing.

Compiled for Anand's Xeon 6 agentic AI infrastructure reference set — cross-references the Storage Architecture, Networking, and Memory Technology references and the Xeon 6 SKU workbook. Verify SN50 absolute TFLOPS and HBM bandwidth against an official spec sheet before using in procurement-facing documents, as these remain press-derived estimates as of Aug 2026.